

ABSTRACT

This study explores the impact of AI tool adoption on job performance, satisfaction, and job security among SME employees.

Using surveys and interviews, we found that higher AI proficiency correlates with increased efficiency and reduced errors, though non-technical users benefit most.

A section of this study explores SME employees' perceptions of AI in creative roles. Findings reveal both benefits, like efficiency gains, and concerns, such as job security risks and creativity constraints. The research highlights a need for balanced AI use that preserves human artistry while enhancing productivity.

Transparent, balanced AI usage is critical to maintaining employee trust and morale, offering SMEs guidance for ethical and effective AI integration.

INTRODUCTION

This study examines how third-party artificial intelligence (AI) tools relate to employee job performance, satisfaction, and perceptions of job security in small and medium-sized enterprises (SMEs).

As SMEs adopt AI technologies to improve efficiency and productivity, it is crucial to understand the implications for employees, particularly regarding how these tools enhance or challenge their work experiences. This study aims to fill the gap in understanding the specific effects of AI integration on employees in these resource-constrained environments.

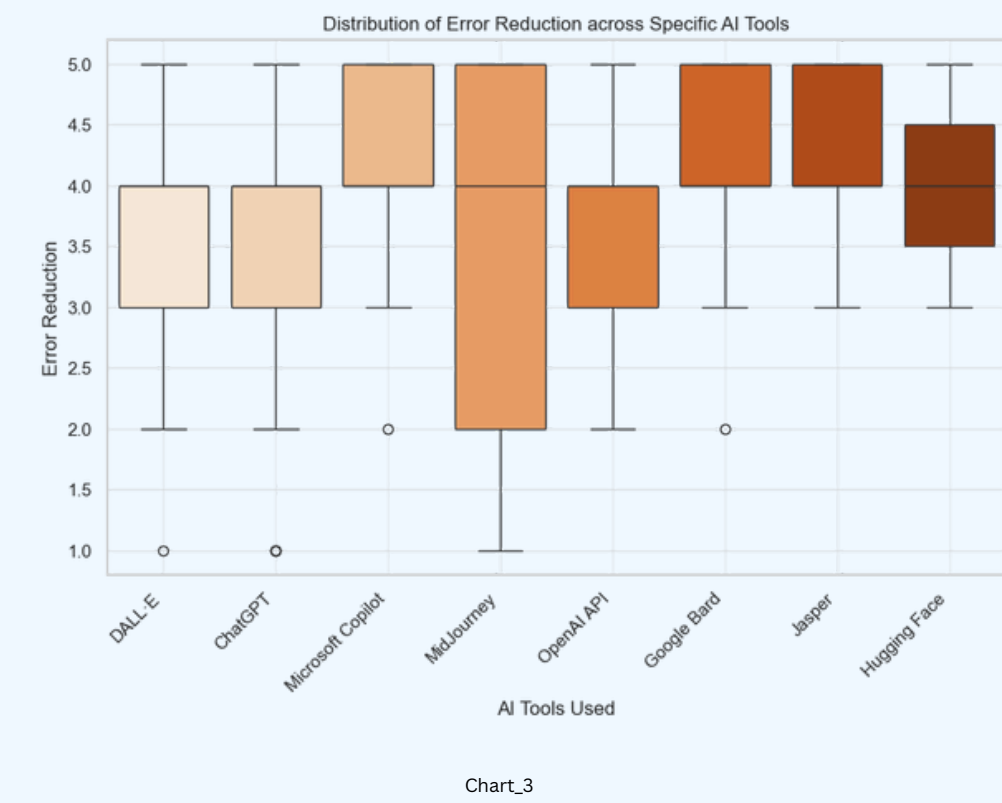
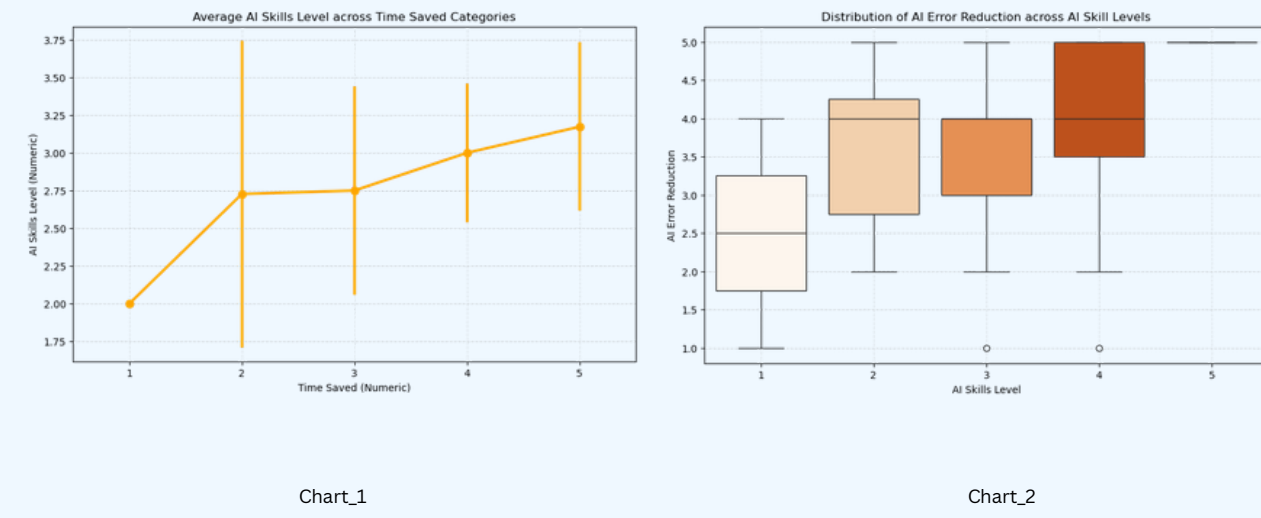
The research focuses on AI's role in enhancing operational efficiency, its application across different business sizes, and the relationship between AI skills and employees' sense of job security. Additionally, it investigates employee perspectives in creative fields, where AI intersects with unique job demands and skills.

By addressing these areas, this research aims to provide a comprehensive understanding of AI's diverse impacts within SME environments.

RESULTS

Does AI Play a Role in Operational Efficiency?

- Charts 1 and 2 indicate a positive correlation between AI skill levels and both time savings and error reduction. Users with higher AI skills tend to experience greater time savings and improved performance or error reduction compared to users with lower AI skills.



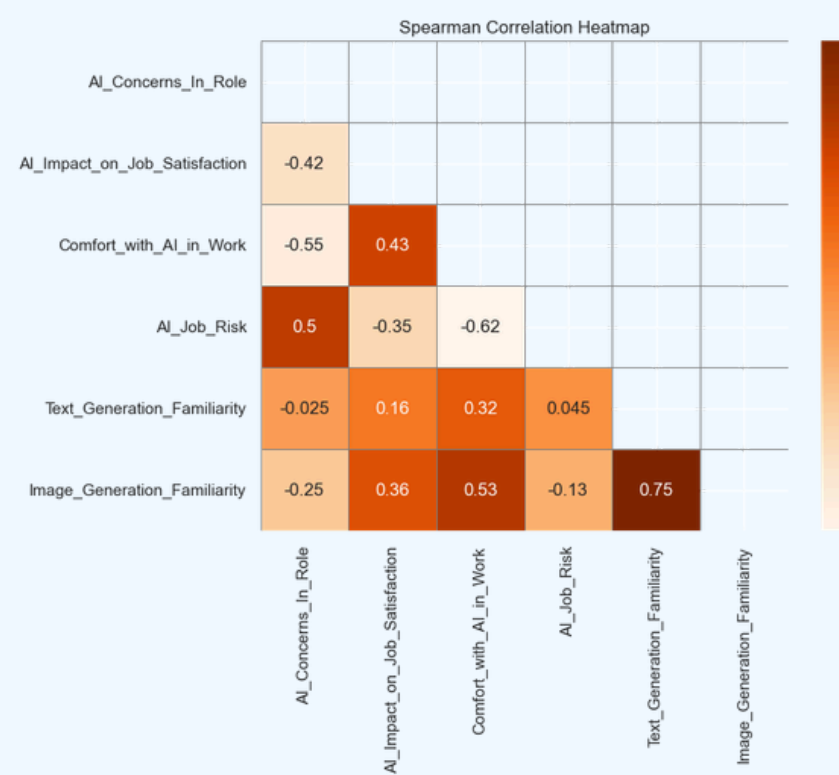
- Chart 3 shows that Google Bard and Jasper provide a more consistent perception of error reduction across users. MidJourney achieves the highest median perception of error reduction but varies widely in effectiveness. ChatGPT and DALL-E exhibit a broad range in perceived effectiveness, with some users reporting minimal benefits in error reduction.

How do employees perceive the implementation of AI in creative fields?

- AI as a Tool: Employees see AI primarily as a means to improve efficiency in technical tasks, not as a source of creativity.

- Job Satisfaction Impact: AI reduces workloads but can lower professional pride and satisfaction, leading to a disconnect from work.

- Job Security Concerns: About 70% of participants worry about AI threatening their job security, with concerns linked to discomfort in using AI.



- Trust and Acceptance: Transparency and training increase trust in AI; skepticism arises from a lack of understanding of its purpose.
- Ethical Concerns: Issues of originality and reliance on AI highlight the need for regulations to safeguard human artistry.
- Preferred Integration: Employees advocate for a balanced approach to AI that supports rather than replaces human creativity.
- Skill Development Issues: Over-reliance on AI for technical tasks may impede skill development, raising concerns about expertise stagnation.
- Management's Role: Support for AI integration differs among managers; those with creative backgrounds value human creativity more than efficiency, causing potential tensions.

AFFILIATES



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METHODOLOGY

Mixed-Methods Approach:

- Combines quantitative surveys and qualitative interviews on AI's impact in SMEs.
- Qualtric surveys for SME employees on AI usage, job satisfaction, and performance, administered via Prolific.
- Semi-structured interviews to explore AI's impact on job performance and satisfaction, conducted via Teams.

Data Security and Anonymity:

- Data will be anonymized and stored securely; unique identification numbers will protect participant confidentiality.

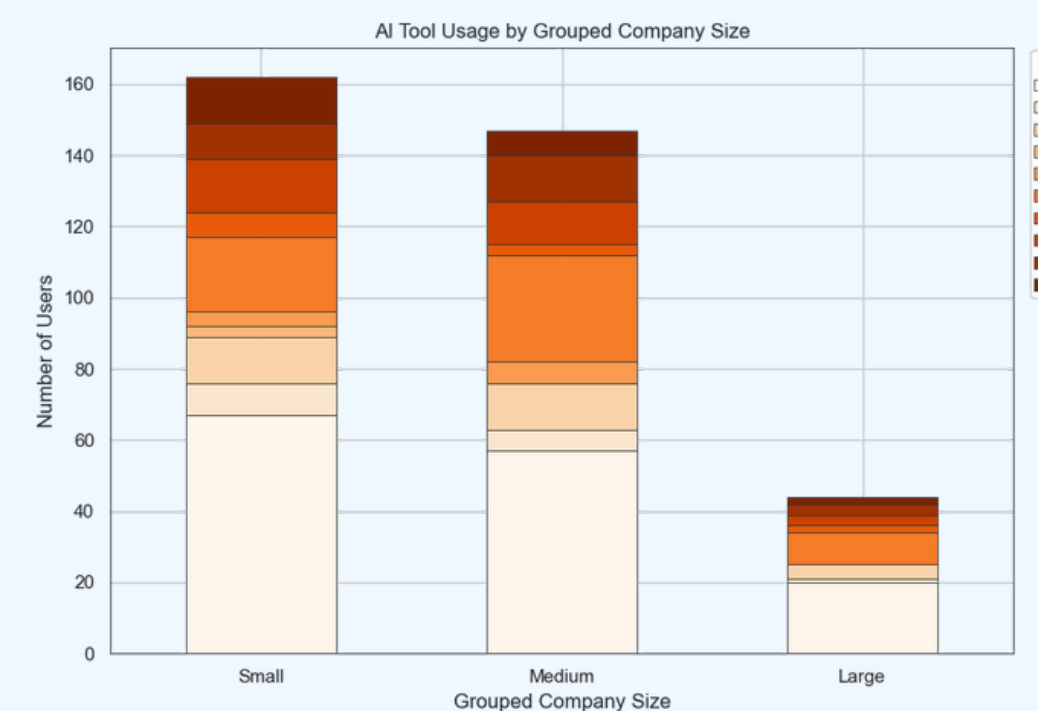
Data Analysis:

- Responses analyzed using descriptive and inferential statistics, with thematic analysis for interview transcripts.

Ethical Considerations:

- Informed consent required from participants; study reviewed by supervisors for ethical compliance.

How does the frequency of AI tool usage at work vary across businesses of different sizes?



- ChatGPT is widely used across company sizes, with 41.4% usage in smaller companies and peaking at 45.5% in larger ones, highlighting its broad appeal.

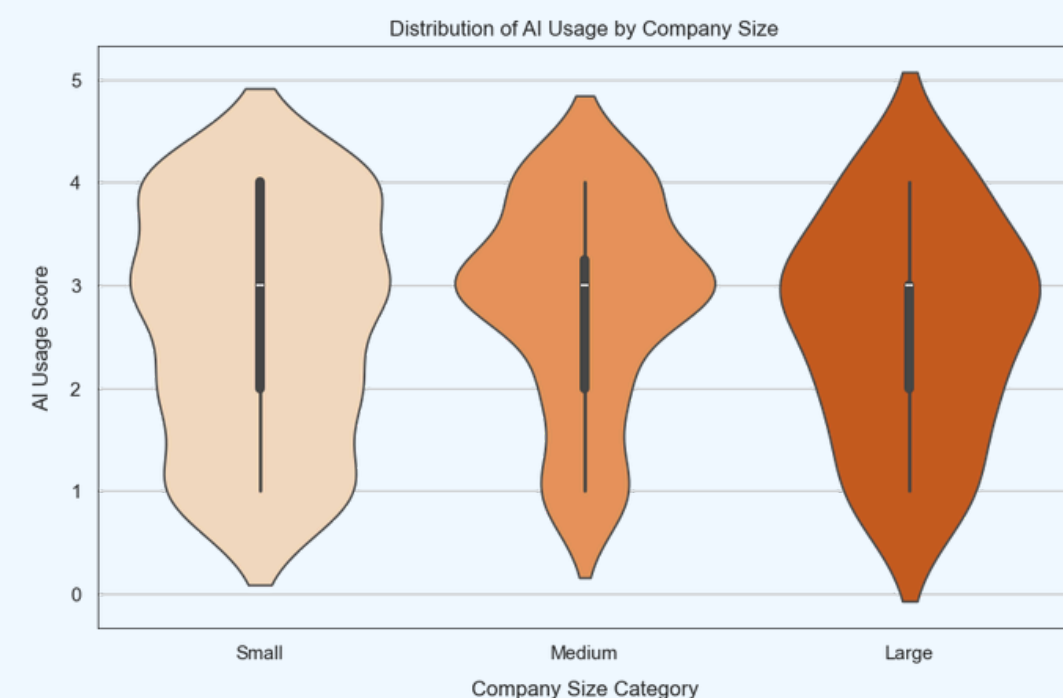
- Microsoft Copilot adoption rises to 20.4% in medium-sized companies, reflecting a growing preference for productivity-enhancing tools.

- DALL-E and Hugging Face have lower usage rates, with DALL-E at just 5.6% and Hugging Face not utilized in larger companies, indicating a trend toward more established AI solutions.

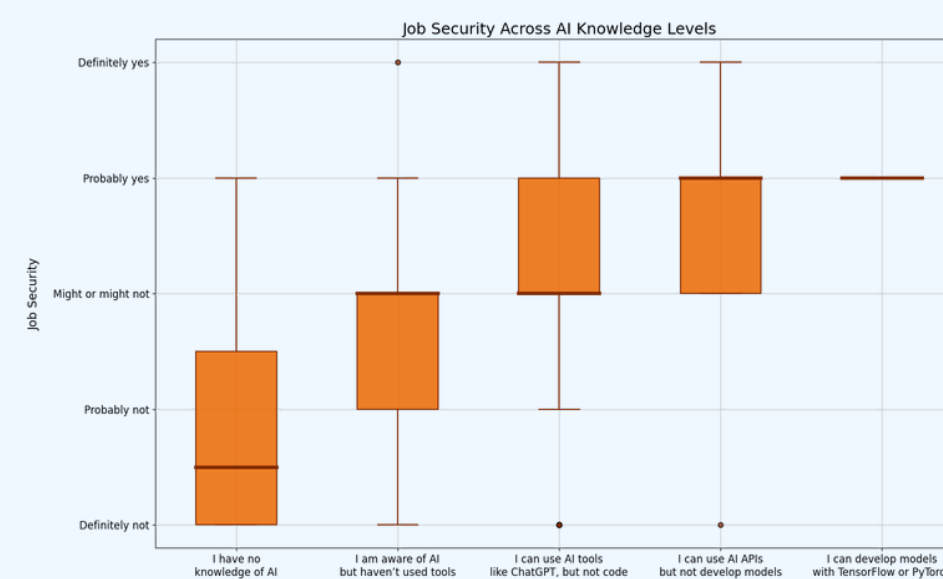
- AI usage is consistent among various company sizes, however medium sized businesses reported a slightly higher occasional usage of AI.

- Challenges the assumption that larger businesses use AI tools more consistently due to greater resources.

- Existing Literature suggests that organizational culture and leadership support may be more critical than resource allocation in influencing AI adoption, highlighting the need for businesses to cultivate an innovative environment.



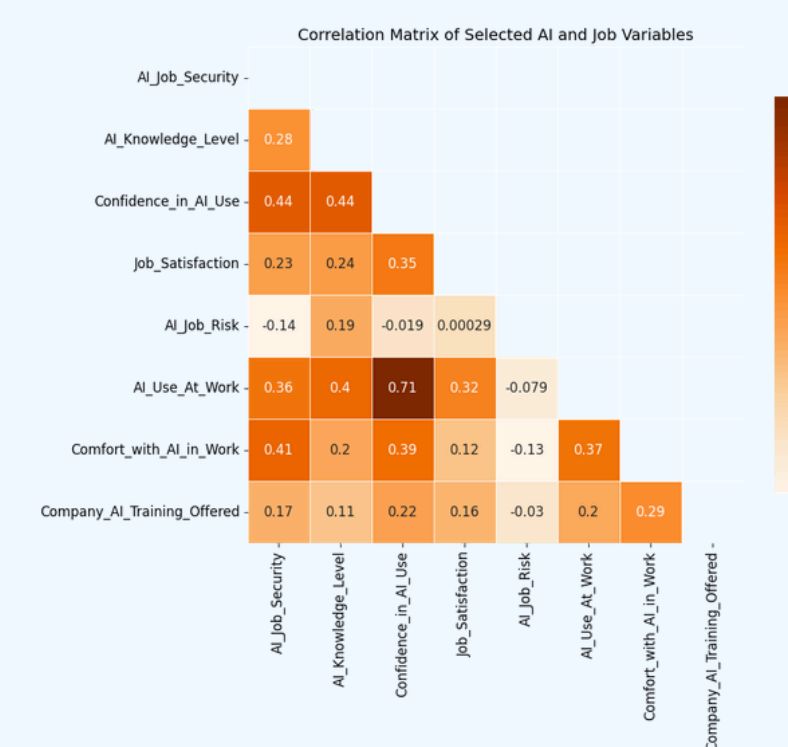
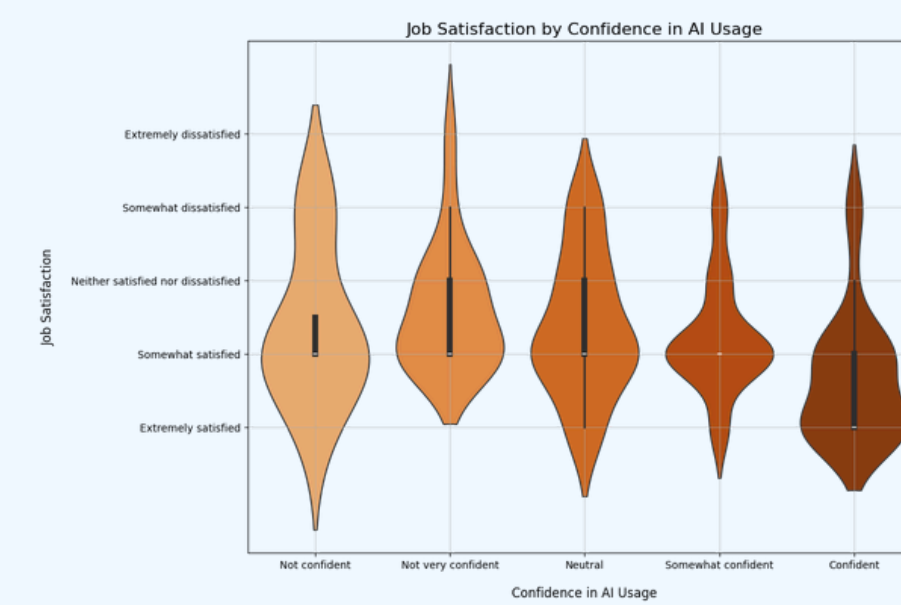
How does having the skills to use AI tools relate to SME employees' perceptions of job security?



- As AI knowledge increases, median job security rises, suggesting that employees with better AI skills perceive their abilities as contributing positively to job stability. Greater knowledge of AI use is linked to higher confidence in using AI among SME employees, which has also been shown to positively impact their job security.

- Additionally, as confidence in AI increases, job satisfaction scores rise, indicating that SME employees who feel confident in using AI are more likely to be satisfied with their jobs.

- A strong positive correlation (0.71) between Confidence in AI Use and AI Use at Work suggests that employees who are more confident in their AI skills are more likely to use AI tools in their roles.



DISCUSSION

These findings suggest that increasing AI skills and confidence in AI usage among employees can enhance perceptions of job security and job satisfaction, reinforcing the value of AI in the modern workplace. However, employees with advanced AI skills also recognize potential job risks associated with AI implementation, underscoring the need for companies to address these concerns as they adopt more AI tools.

This study found no significant difference in AI tool usage frequency among small, medium, and large businesses, suggesting that resource availability alone doesn't drive AI adoption. Instead, organizational culture and leadership may play a stronger role in influencing AI usage. These results encourage businesses to prioritize a culture of innovation over solely expanding resources for AI integration.

The study reveals a nuanced view of AI in creative fields: while AI enhances task efficiency, employees feel it lacks the depth required for creative inspiration, potentially diminishing job satisfaction. Concerns about job security, ethical implications, and creative authenticity are prevalent, highlighting the need for balanced AI integration that respects human creativity and prioritizes transparency and trust.

The study suggests that higher AI skills are linked to greater time savings, highlighting skill development as a driver of productivity, especially for less technical users. For error reduction, tool choice is more influential than skill level: reliable tools like Google Bard and Jasper minimize errors, while flexible options like ChatGPT and MidJourney show more variable results.

CONCLUSIONS

Employees with AI skills tend to perceive greater job security, with higher confidence in AI usage positively correlating with both job satisfaction and a stable sense of their roles in SMEs.

Factors like culture and leadership play a crucial role, suggesting that fostering innovation may be more impactful than resource allocation alone. Future studies should examine industry-specific factors and leadership attitudes to gain deeper insights into effective AI integration.

Employees appreciate AI's efficiency but worry about its impact on creativity, job security, and job satisfaction. They call for transparent AI use and supportive management to foster trust. Overall, employees favor AI as a supportive tool, not a replacement, to preserve the human element in creativity.

The study suggests that enhancing AI skills boosts time savings, particularly benefiting non-technical users, while tool choice is essential for error reduction. Reliable tools like Google Bard and Jasper ensure accuracy, whereas flexible tools like ChatGPT and MidJourney show varied results.

REFERENCES

- G. Amato, M. Behrmann, F. Bimbot, et al., AI in the media and creative industries. (May 2019).
- Banholzer, M., Fletcher, B., LaBerge, L., & McClain, J. (2023, August 31). Companies with innovative cultures have a big edge with generative AI. McKinsey & Company.
- Ebuka, A. A., Emmanuel, D., & Idigo, P. (2023). Artificial Intelligence as a catalyst for the Sustainability of Small and Medium Scale Businesses (SMEs) in Nigeria. Annals of Management and Organization Research, 5(1), 1–11.
- Cubric, M. (2020). Drivers, barriers and social considerations for ai adoption in business and management: A tertiary study. Technology in Society, 62:101257.